

ADARSH KATARE

SOFTWARE DEVELOPER

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[LinkedIn](#) | [GitHub](#) | [Leetcode](#) | [CodeChef](#) | [Portfolio](#) | [Codeforces](#)

EDUCATION

Indian Institute of Information Technology Bhopal

September 2023- June 2027

Bachelor of Technology - Electronics and Communication Engineering

Experienced In: **DSA, OOPS, Operating System, Computer Network, Database Management System**

PROFESSIONAL EXPERIENCE

Machine Learning Research Internship - IIT BHU – Varanasi

May 2025 - July 2025

- Constructed joint angle prediction pipeline for 6 DOF robot arm by configuring **ES-RNN and ANN** architectures; applied sequential feature extraction and residual connections, improving prediction **accuracy by 80%**
- Benchmarked **Random Forest, XGBoost, and SVM** on 300K+ records, identifying optimal model with **19%** higher accuracy than baseline

AI & MERN Developer Internship - HYPWEB SOLUTIONS LLP - Remote

March 2025 - April 2025

- Engineered a Linux command correction chatbot by **finetuning Meta LLaMA3-8B LLM** using **Low Rank Adaptation (LoRA)** technique achieving **89.4% accuracy** across 2000+ diverse user queries
- Integrated a **REST API** layer using **Node.js and Express.js** to serve the chatbot, **handling 500+ queries per day** with structured query logging in MongoDB reducing latency up to 60%

PROJECTS

HealthifyAi | Python, LLM, Finetuning, Flask, React, MongoDB, JWT

[Demo](#) | [Github](#)

- Architected an **AI powered food product analysis pipeline** leveraging **LoRA finetuning** on **Mistral-7B LLM** with a dataset containing 15K+ samples, attaining transformer **accuracy up to 86%**
- Implemented a multimodal workflow integrating **OCR, NLP normalization** boosting **text accuracy by ~35%**, and **ngrok** for exposing deployment ready API key, reducing model latency by **30%**.
- Deployed a **full stack web application** on Vercel & Render with **CI/CD** using React, Flask, and MongoDB Atlas with JWT authentication, reducing unauthorized API access by **100%**

Sign Language AI | OpenCV, TensorFlow, CNN-LSTM

[Demo](#) | [Github](#)

- Built a **sign language recognition system** using **CNN-LSTM architecture** for automated object tracking and motion analysis workflow using **OpenCV**, achieving **94% accuracy** across image and video datasets
- Achieved **~85% classification accuracy** on custom gesture dataset while optimising prediction latency by **~19%**, enabling consistent **real-time response** under time constrained environment
- Designed unified training pipeline (**Xception, EfficientNetB4, ResNet50, CNN-RNN**); deployed on Hugging Face Spaces with **Streamlit UI and Docker** containerization

ACHIEVEMENTS

- Secured **2nd Runner-up** at Smart India Hackathon, recognised for building a real-world scalable tech solution
- Acquired **Knight Badge** and ranked in the **top 5.51 %** globally on LeetCode with a peak score of **1864**
- Accomplished a **global rank 764** among 35,000+ participants in the CodeChef Starters Contest; earned **3 Star** rating with a peak **rating of 1697**
- Attained **Specialist rank on Codeforces**, with highest rating of **1427**, securing rank under **2800 in Div2** contest
- Secured **1st Rank in District** in 12th Board CBSE examination with **93.4%**; scored **92.6%** in 10th Board CBSE

SKILLS SUMMARY

Programming Languages: Python, C/C++, JavaScript (ES6), TypeScript, SQL

Frontend & Backend: React.js, HTML5, CSS3, Node.js, Express.js, FastAPI, REST API, Flask, Microservices

AI/ML and Automation: Pytorch, TensorFlow, OpenCV, Pandas, Scikit Learn, LLM, RAG, LangChain

Tools & DevOps: Git, GitHub, Docker, MongoDB, PostgreSQL, MySQL, Postman, Hugging Face, Linux

ADDITIONAL INFORMATION

Computer Vision Lead | SPARK Club, IIIT BHOPAL

November 2025- Present

- Orchestrated and led Computer Vision workshop for **150+** students, imparting hands-on skills in **OpenCV, image processing, and YOLO based object detection**, while mentoring a team of 4 machine learning developers, fostering a collaborative learning environment and tackling complex challenges under pressure